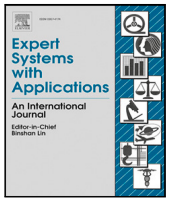
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How to identify pollen like a palynologist: A prior knowledge-guided deep feature learning for real-world pollen classification

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ABSTRACT

Airborne pollen identification is crucial to help patients prevent pollinosis symptoms. Existing data-driven methods rely on large-scale pollen images with simple backgrounds. In real scenarios, the background is complex and the data scale is small. Therefore, these methods suffer from two challenges: (1) Irrelevant information interference; (2) Incomplete feature attention. To overcome these challenges, we propose a prior knowledge-guided deep feature learning (PK-DFL) for real-world optical microscope image classification. Its main steps are as follows: **Pollen location** is designed to locate pollen grains based on color features, aiming to boost the accuracy of shape and texture prior feature extraction. **Shape-texture awareness** helps to extract the shape and texture of pollen grains via predefined feature extractors (i.e., a set of shape descriptors and an improved SFTA). These features are used to construct two types of prior knowledge, namely shape-texture attention maps (STA maps) and shape-texture feature vectors (STF vectors). **Pollen classification** uses a deep network (CNN) to classify pollen via imitating the pollen identification procedure of palynologists. It uses STA maps to weight pollen images and convolutional feature maps for instructing the CNN to focus on critical areas of pollen images (**for the first challenge**). STF vectors are employed to obtain the inter-class similarity of pollen via template matching. This information is further converted to soft targets that are used to supervise the CNN attending to comprehensive key features (**for the second challenge**). Extensive experiments on real-world datasets demonstrate the effectiveness of our PK-DFL (with accuracy and F1-score over 88%).

1. Introduction

Allergenic pollen, the predominant aeroallergen, always induces a series of pollinosis symptoms (rhinitis, asthma and conjunctivitis) in patients (Fernstrom & Goldblatt, 2013; Ribeiro et al., 2013). These allergic diseases are severely harmful to human health or even life-threatening. More than 10% of adults and 40% of children worldwide suffer from pollen allergy (Bieber et al., 2016; Reitsma, Subramaniam, Fokkens, & Wang, 2018). Medical specialists suggest that one of the effective measures to prevent allergy symptoms is to avoid pollen exposure (Sur & Plesa, 2015). The allergenic capacities of pollen are distinct, and the personal sensitivity is specific to each species (Rodriguez-Damian, Cernadas, Formella, Fernández-Delgado, & De Sa-Otero, 2006). Therefore, the classification of airborne allergenic pollen is a key step in the treatment of pollinosis. Pollen identification is gradually being promoted in many countries around the world (Buters et al., 2018). The traditional method is to observe and identify pollen grains in slides by a qualified palynologist under an optical microscope (OM) (Khanzhina et al., 2022). However, the manual identification method is time-consuming, laborious, and mostly subjective, making it difficult to provide real-time and accurate pollen forecast. Consequently, developing an automatic pollen classification method with high accuracy and efficiency is required eagerly.

Several studies have been done for pollen automatic identification. These methods can basically be divided into two types: model-driven methods and data-driven ones. The model-driven approach extracts predefined features such as shape and texture of pollen grain, followed by constructing traditional classifiers to identify pollen categories (Almotairi, 2019; Chen et al., 2006; del Pozo-Baños et al., 2012; Faegri, Kaland, Krzywinski, et al., 1989; Filipovych, Daood, Ribeiro, & Bush, 2016; Goncalves et al., 2016; Marcos et al., 2015; Punt, Hoen, Blackmore, Nilsson, & Le Thomas, 2007; Ranzato et al., 2007; Rodriguez-Damian et al., 2006; Tcheng, Nayak, Fowlkes, & Punyasena,

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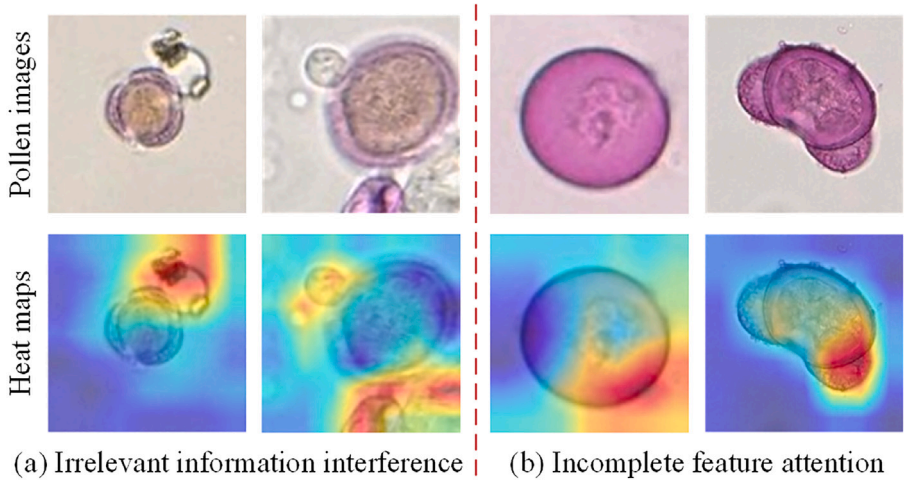


Fig. 1. Examples of OM pollen images and visualization results of the baseline.

2016; Tello-Mijares & Flores, 2016; Travieso, Briceño, Ticay-Rivas, & Alonso, 2011; Wang, Bao, Lin, & Wang, 2019; Zhang, Fountain, Hodgson, Flenley, & Gunetileke, 2004). Specifically, Rodriguez-Damian et al. (2006) used three classifiers (i.e., minimum distance classifier, multi-layer perception, support vector machine) to identify the extracted shape and texture features. They employed statistical moments, Fourier descriptors and geometrical information to represent the pollen shape. Grey-Level Co-occurrence Matrix (GLCM), Gray Level Run Length Statistics (GLRLS) and Neighboring Gray Level Dependence Statistics (NGLDS) were used to extract texture. Chen et al. (2006) first calculated geometrical features (i.e., area, perimeter, etc.), first-order statistics and pore/colpus features to describe pollen grains. Then a k-nearest neighbor and a linear classifier were trained to predict pollen categories. Marcos et al. (2015) trained a k-nearest neighbor based on texture, which is calculated using GLCM, Log-Gabor Filter and Discrete Tchebichef Moment. These methods have low accuracy and poor generalization ability, since they only consider handcrafted features and ignore high-level features of pollen images.

The data-driven approach relies mainly on convolutional neural networks (CNNs) to automatically extract discriminable features from images (Bourel et al., 2020; Daood, Ribeiro, & Bush, 2016; de Geus, Barcelos, Batista, & da Silva, 2019; Gui, Wang, Zhu, Zhu, & Zhao, 2021; Holt, Allen, Hodgson, Marsland, & Flenley, 2011; Khanzhina et al., 2022; Khanzhina, Putin, Filchenkov, & Zamyatina, 2018; Mahbod, Schaefer, Ecker, & Ellinger, 2021; Menad, Ben-Naoum, & Amine, 2019; Mercado, Pineda, et al., 2020; Murkute, 2021; Sevillano & Aznarte, 2018; Sevillano, Holt, & Aznarte, 2020; Trenta, Ortis, & Battiato, 2021; Viertel & Konig, 2021). Specifically, Daood et al. (2016), Khanzhina et al. (2018) and Menad et al. (2019) proposed a novel CNN structure for pollen identification, respectively. de Geus et al. (2019), Sevillano and Aznarte (2018) and Sevillano et al. (2020) used classical CNNs (i.e., AlexNet, InceptionV3, ResNet, and DenseNet, etc.) to classify pollen images. Gui et al. (2021) proposed two data augmentation operations to help the deep network focus on the surrounding of pollen grains. For predict pollen categories, an existing fine-grained classification method and the training data augmentation proposed in Gui et al. (2021) were applied by Trenta et al. (2021). These researches eliminate the need to design predefined features and achieve end-to-end training and testing. Such approaches have achieved better performance than model-driven methods thus gradually becoming the mainstream. However, these methods only achieve superior identification accuracy on pollen images with simple backgrounds, and their performance is heavily dependent on large-scale data.

Pollen images in real-world have the following characteristics: (1) **Complex image background.** Due to the specificity of pollen sample collection process, pollen images contain plenty of irrelevant information such as impurities (including animal remains, plant debris, dust, etc.) and bubbles. (2) **Small data scale.** It is difficult to obtain large-scale data because pollen sample collection is limited by geographic location and temporal distribution, and the process of digitizing physical samples is time-consuming and laborious. Therefore, existing data-driven methods face the following two challenges (as shown in Fig. 1) :

Challenge 1: Irrelevant information interference. Pollen images contain extensive irrelevant information such as impurities and bubbles. Experts focus only on pollen grains and ignore unrelated areas when identifying pollen images. Existing CNN-based pollen identification methods can achieve satisfactory performance only on laboratory data. The images in the lab scene contain only pollen grains and are free of other irrelevant targets. In real-world scenarios, existing methods are disturbed by irrelevant information, leading to poor identification performance. Therefore, how to facilitate the deep network to focus on key regions of the image is a crucial issue worth investigating.

Challenge 2: Incomplete feature attention. Pollen grains have various key features such as shape, texture, and aperture. Experts often combine multiple features to accurately determine the category when identifying pollen images. With limited pollen samples, existing data-driven deep models are difficult to focus fully and accurately on important features. Attending to single or partial features leads to pollen misclassification. Therefore how to improve the attention of deep models to key features is an important challenge urgently to be addressed.

We solve the above challenges based on expert pollen identification basis and process (see Fig. 2). In terms of classification basis, since shape and texture are the primary and important features for palynologists to identify pollen grains, we extract these two features as prior knowledge to guide the deep network. In the process of pollen identification, the expert first focus on crucial areas (pollen grains) and then attend to key features (i.e., shape, texture, etc.). For step 1, the expert picks pollen grains based on features such as shape and texture, rather than concentrating on unrelated background information. This is consistent with the attention mechanism in computer vision tasks. Therefore, we construct attention maps based on shape and texture features to instruct the deep network to focus on key regions of pollen images (**solving challenge 1 by Attention Instruction**). For step 2, given the pollen image to be identified, the palynologist combines multiple discriminative features to match with the target pollen class for obtaining the pollen inter-class similarity. The target pollen with the highest similarity is the predicted category. This inter-class similarity knowledge reacted to the computer vision task is the soft target in knowledge distillation. Therefore, we extract shape and texture to

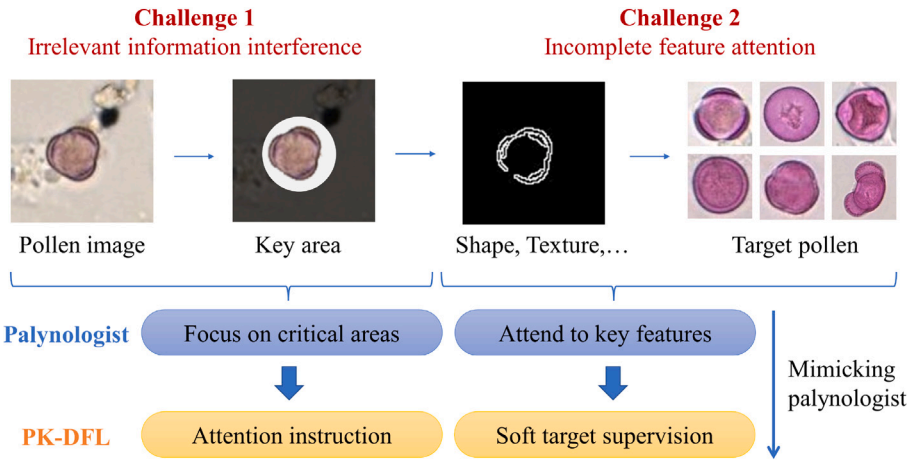


Fig. 2. The identification basis and process when experts classify pollen.

build soft targets that are used to supervise deep networks focusing on comprehensive pollen grain features by knowledge distillation (**solving challenge 2 by Soft Target Supervision**).

Inspired by the above instruction of domain knowledge, we assume that imitating the pollen identification basis and process of palynologists can effectively improve the performance of pollen classification models. Namely, shape and texture prior features can help deep networks avoid irrelevant object interference and focus on more complete pollen features. This can further help the classification network to improve the accuracy of pollen identification. Specifically, the texture of the pollen is regular, while that of the impurities is the opposite. In addition, the shape and texture of pollen grains are different. For example, the key features of Cupressaceae are round and the contents in the center of pollen grains; the primary identification basis of Artemisia is tri-lobed shape. Therefore, shape and texture prior features are helpful for pollen identification.

Based on the above assumptions, this paper proposes a prior knowledge-guided deep feature learning (PK-DFL) for real-world allergenic pollen image classification. Specifically, we first design a pollen location module to obtain pure pollen images, aiming to help the next module extract precise shape and texture features. Considering color features, this module locates pollen grains in pollen images using Digital Image Processing technology (e.g., image filter, morphological operation). Then, a shape-texture awareness module is developed to extract the shape and texture of pollen grains. We design a set of shape descriptors to describe pollen shape and an improved SFTA algorithm to extract texture. These features are further devoted to the next module as two types of prior knowledge (i.e., STA maps and STF vectors). Finally, guided by prior knowledge, the pollen classification module learns high-level features of pollen images using a CNN to predict pollen categories. The STA map is weighted to the pollen image and the convolutional feature map, aiming to guide the classification network to avoid the interference of irrelevant information and focus on the key areas of the pollen image. We employ STF vectors to get the inter-class similarity through template matching. This information is further turned into soft targets which are used to guide the deep network to focus on more comprehensive key features of pollen grains through knowledge distillation.

Overall, our contributions can be summarized as follows:

- We propose a prior knowledge-guided deep feature learning network for real-world pollen image classification. After locating pollen grains based on color features, we design predefined feature extractors (a set of shape descriptors and an improved SFTA algorithm) to extract the shape and texture of pollen grains. These features serve as prior knowledge to guide the pollen identification process of the deep network.
- We construct STA maps based on the shape and texture. These maps are weighted onto pollen images and different convolutional feature maps to instruct the deep network to focus on key regions of the pollen image. Thus we solve the problem of irrelevant information interference (**challenge 1**).
- Considering shape and texture, we build STF vectors that are used to obtained inter-class similarity through template matching. This information is further transformed into soft targets. Knowledge distillation is used to guide the network to learn comprehensive key features of pollen grains. Thus the problem of insufficient attention to key features is effectively addressed (**challenge 2**).

Extensive experiments are conducted on real-world datasets, including comparison experiments and ablation experiments. We perform qualitative and quantitative evaluations and experimental results show that our PK-DFL achieves remarkable classification performances with 88.96% accuracy and 88.20% weighted F1-score.

The remainder of this paper is organized as follows. The proposed PK-DFL is introduced in Section 2. Experimental settings and results on real-world datasets are shown in Section 3. Section 4 concludes this paper and suggests future works.

2. Method

In this section, we introduce the proposed PK-DFL in detail. Our key insight is to mimic the basis and process of expert pollen identification. Shape and texture prior knowledge is used to guide the classification network to focus on key regions of the image and improve the attention of key features. As shown in Fig. 3, our PK-DFL mainly contains 3 modules, namely Pollen Location, Shape-Texture Awareness and Pollen Classification. Firstly, the pollen location module uses digital image processing methods to locate pollen grains based on color features facilitating accurate extraction of prior features. Then the shape-texture awareness module designs shape descriptors and an improved SFTA algorithm to extract the shape and texture of the pollen grain. These features are further transformed into STA maps and STF vectors to serve the next module. Finally, the pollen classification module uses a CNN guided by prior knowledge to learn high-level features of pollen images for predicting pollen categories. The STA map is weighted to the original pollen image and the convolution feature map for instructing the classification network to focus on critical areas of the pollen image. Based on the STF vector, we use template matching to get the inter-class similarity of pollen. This information is further converted to soft labels that supervise the deep network attending to comprehensive pollen key features.

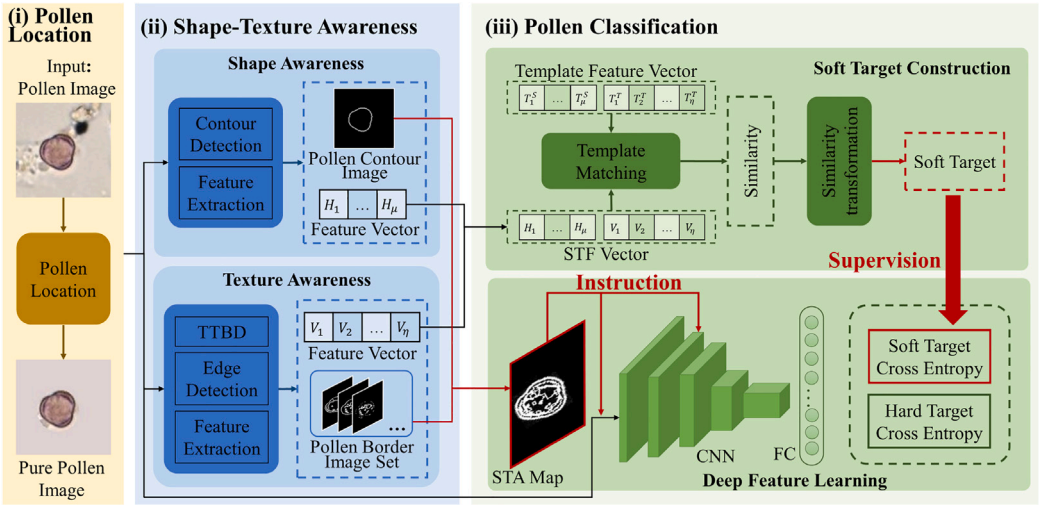


Fig. 3. The overall framework of our PK-DFL.

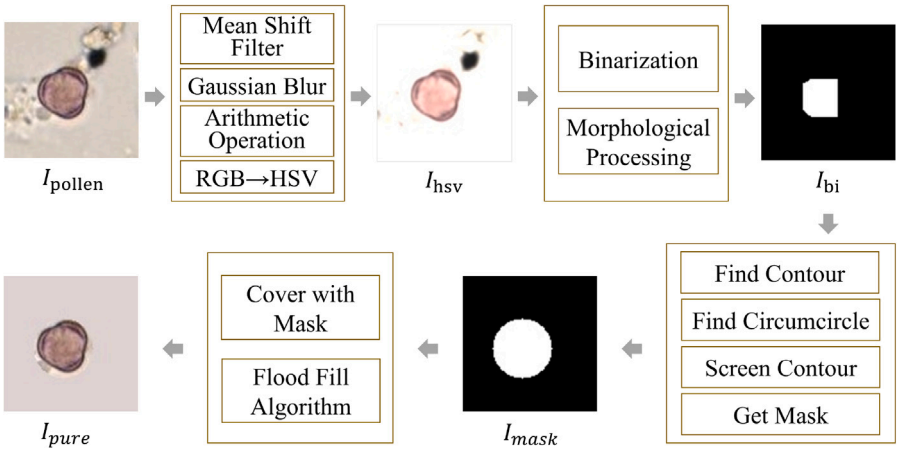


Fig. 4. The pipeline used to locate pollen grains.

2.1. Pollen location module

In PK-DFL, a pollen location module is first designed to generate pure pollen images used in the next module. Pollen samples collected are usually colored to better identify pollen grains. In this case, pollen grains and unrelated targets in the pollen digital image usually show different colors (pollen grains are purple, while bubbles and impurities are usually not that). Hence, we locate pollen grains based on color features to help the next module extract accurate shape and texture prior features. The specific process is shown in Fig. 4.

A mean shift filter and a 3×3 kernel Gaussian filter are first applied to reduce background noise. We multiply the image by a constant greater than 1 to increase the brightness of the image. After changing the color space from RGB to HSV, we binarize the image I_{hsv} based on the color threshold. To fill small closed regions, we perform a closing operation with a kernel of 30×30. An opening operation with a kernel of 20×20 is applied to remove noise outside the object.

Then we detect contours present on the binary image I_{bi} and calculate the circumcircle of the contour. These contours are filtered according to the radius of the circumcircle to remove small impurities with similar color to pollen grains. We construct the pollen mask I_{mask} based on the contour circumcircle. After the original pollen image is covered with the mask, we employ the flood filling algorithm to make the image similar to the original ones in the background color. Finally, we get the pure pollen image I_{pure} which helps the next module to extract accurate prior knowledge.

2.2. Shape-texture awareness module

According to expert knowledge, the shape and texture are the key features to identify pollen grains. Therefore, we design a shape-texture awareness module to extract the shape and texture of pollen grains. These features are used to guide the pollen classification module as prior knowledge.

2.2.1. Shape awareness

The shape is the basic characteristic of pollen grains, and the shape of each species is specific (e.g., the Cupressaceae is round, the Artemisia is tri-lobed, while the Pinaceae is oval). Using this knowledge, we design a set of shape descriptors to extract shape features from pollen contour images. Specifically, we first detect the edge of pollen grains based on Canny operators. The *findContours* in the OpenCV library is then used to extract outer contours of pollen. Finally, shape descriptors are designed to describe pollen shape. Our shape descriptors include geometric features and Hu moments (Hu, 1962). Geometric features can describe the size and basic shape of pollen grains, containing following metrics.

- RAP: The ratio between the area and the perimeter of the pollen grain.

$$RAP = \frac{A}{Pe} \tag{1}$$

- Compactness: Measures how circular the pollen is.

$$Compactness = \frac{Pe^2}{4\pi A} \tag{2}$$

- Rates of changes: The measures of D_{max}, D_{min}, and D_{mean}, which are defined as the maximum, the minimum, and the average distance between the centroid and the boundary of the pollen.

$$RoC_1 = \frac{D_{max}}{D_{min}} \tag{3}$$

$$RoC_2 = \frac{D_{max}}{D_{mean}} \tag{4}$$

$$RoC_3 = \frac{D_{min}}{D_{mean}} \tag{5}$$

- Eccentricity: This metric, reflecting elongation, estimates the ratio of the major axis and the minor axis of the best fit ellipse over the object.

$$Eccentricity = \frac{D_{ma}}{D_{mi}} \tag{6}$$

- RC: This metric is related to the roundness and circularity.

$$RC = \frac{Pe - \sqrt{Pe^2 - 4\pi A}}{Pe + \sqrt{Pe^2 - 4\pi A}} \tag{7}$$

- Fullness: The ratio of the pollen area to bounding rectangle area.

$$Fullness = \frac{A}{A_{br}} \tag{8}$$

- Holes: The contour and shape of pollen are denoted as C and S, respectively. $p_c = (x_c, y_c)$ denotes the centroid of shape S. p_i refers to a point in the contour C. This metric is defined as:

$$Holes = \sum_{i=0}^{Pe-1} D_{max} - \|p_c - p_i\| \tag{9}$$

Hu moments, composed of seven invariant moments, can describe not only the invariant morphological features of shapes but also higher-order statistical features. By calculating these shape descriptors, we obtain a shape feature vector with 16 dimensions.

2.2.2. Texture awareness

Texture, describing the granularity and repetitive patterns of regions in the image, is a key feature for experts to identify pollen grains. We employ an improved SFTA algorithm (Costa, Humpire-Mamani, & Traina, 2012) to extract the texture of pollen. Fig. 5 shows the process of improved SFTA, including 4 steps.

Step 1: Pollen images are converted to grayscale.

Step 2: We employ the TTBD algorithm proposed by Costa et al. (2012) to decompose the grayscale image $I_g(x, y)$ into a set of binary images. Specifically, a multi-level Otsu algorithm (Liao, Chen, Chung, et al., 2001) is used to determine a threshold set T containing n_t (a predefined number) thresholds. Based on the paired thresholds selected from $T \cup \{V_{max}\}$, a two-threshold segmentation (see Formula (10)) is applied to obtain binarized images. V_{max} denotes the maximum possible gray level of the pollen grayscale image.

$$I_{bi}(x, y) = \begin{cases} 1, & if\ tl \le I_g(x, y) \le tu \\ 0, & otherwise \end{cases} \tag{10}$$

where tl denotes lower threshold and tu denotes upper threshold, $tl\ and\ tu\ \in T \cup \{V_{max}\}$. Through the TTBD algorithm, we can get $2n_t - 1$ binary images.

Step 3: We extract the edges of binary images with Formula (11) to obtain border images.

$$I_{bo}(x, y) = \begin{cases} 1, & if\ \exists (x', y') \in N8[(x, y)]: \\ & I_b(x', y') = 0 \wedge I_b(x, y) = 1 \\ 0, & otherwise \end{cases} \tag{11}$$

where N8[(x, y)] is the set of pixels that are 8-connected to (x, y).

Step 4: Three metrics (size $\mathcal{A}$, mean gray level $\mathcal{M}$, and fractal dimension D_H) are calculated based on grayscale images and border images. Different from the original SFTA, we calculate size (Formula (12)) and mean gray level (Formula (13)) based on border images instead of binary images.

$$\mathcal{A} = \sum_{i=1}^{H} \sum_{j=1}^{W} I_{bo}(i, j) \tag{12}$$

$$\mathcal{M} = \frac{1}{\mathcal{A}} Sum(B) \tag{13}$$

$$B = \{ I_{gr}(x, y) | I_{bo}(x, y) = 1 \} \tag{14}$$

where, H and W are the height and width of the border image, respectively. $Sum(B)$ refers to the sum of all elements in the set B. I_{gr} is the grayscale image. Using Hausdorff's dimension, fractional dimensions are defined to quantify the texture complexity of pollen grains. Assuming a border image possesses a Euclidean dimension D_E, its Hausdorff's fractal dimension D_H is defined as follows:

$$D_H = \lim_{\epsilon \to 0} \frac{log N(\epsilon)}{log \epsilon^{-1}} \tag{15}$$

where $N(\epsilon)$ is the number of hyper-cubes in dimension D_E and ϵ is the length used to fill objects in the border image. Based on the improved SFTA, we can get a texture feature vector with a length of $3(2n_t - 1)$.

2.2.3. STA map and STF vector construction

STA maps: Normalized pollen contour images (generated in the *shape awareness*) and border images (obtained in the *texture awareness*) are fused to obtain STA maps. Inspired by Zheng, Fu, Zha, and Luo (2019), we calculate the average of these images to construct STA maps that takes all shape and texture into consideration. **STF vectors:** We refer to the shape and the texture feature vectors collectively as STF vectors.

2.3. Pollen classification module

The pollen classification module is designed to predict pollen categories under the guidance of prior knowledge obtained in the previous module. We use a classical CNN as the backbone (e.g., VGG19). The manner of prior knowledge guidance mainly includes *attention instruction* and *soft target supervision*. STA maps are used by *attention instruction* to guide the CNN to focus on key areas of the pollen image. Based on STF vectors, *soft target supervision* guides the deep network attending to comprehensive key features.

2.3.1. Attention instruction

Aiming to instruct the classification network focusing on critical regions of pollen images, STA maps are weighted to the input image and the convolutional feature map. We weight the STA map to the pollen image and the feature map of the second convolutional layer according to experiments. Mathematically, the pollen image and the convolutional feature map F can be masked by the STA map M with the following formula:

$$F_{Att} = F \odot \{(1 - \alpha) \cdot M \oplus \alpha\} \tag{16}$$

where, $\odot$ and $\oplus$ represent element-wise multiplication and addition, respectively. α is a threshold to control the influence of STA maps (it is set to 0.4 in this paper).

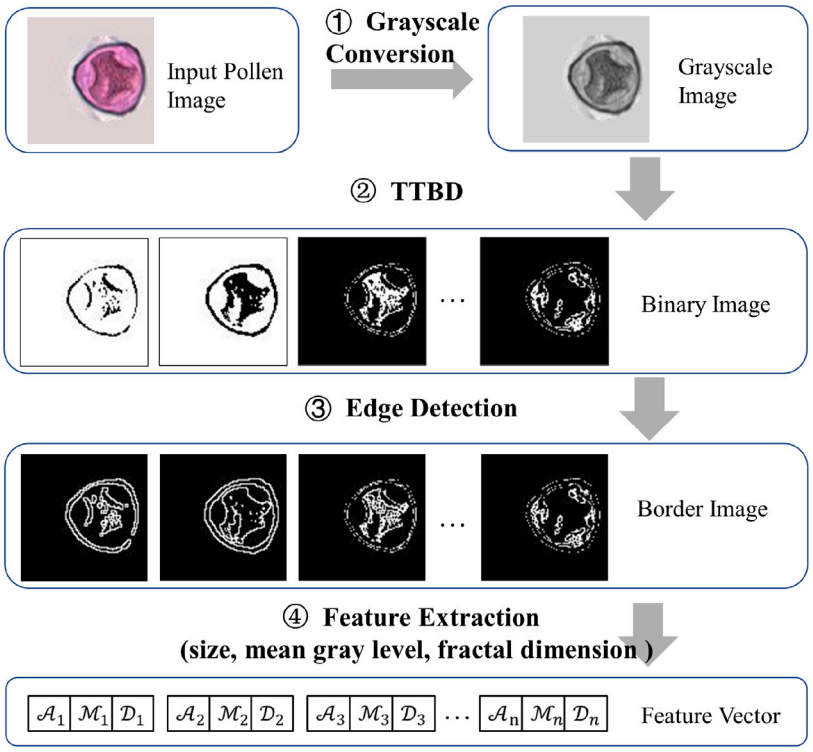


Fig. 5. The process of pollen texture extraction with the improved SFTA.

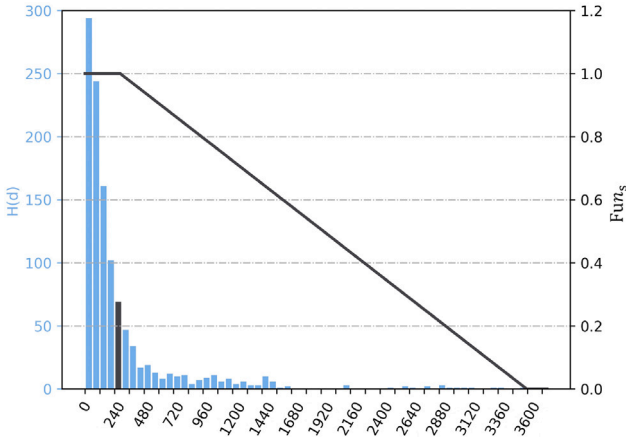


Fig. 6. Distance to the similarity conversion function based on the frequency distribution of shape distances ($\mathcal{U} = 287, \mathcal{V} = 3594$).

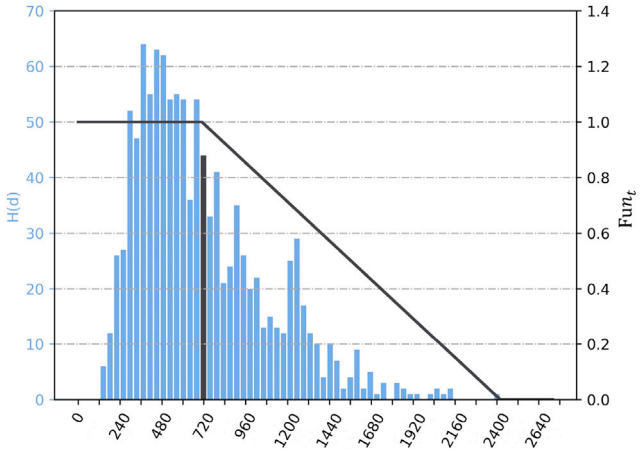


Fig. 7. Distance to the similarity conversion function based on the frequency distribution of texture distances ($\mathcal{U} = 709, \mathcal{V} = 2420$).

2.3.2. Soft target supervision

We use STF vectors to obtain the pollen inter-class similarity by template matching. This information is further converted into soft targets to supervise the classification network focusing on comprehensive pollen key features. This guidance pattern consists of 2 main steps, namely soft target construction and shape-texture knowledge distillation.

Step 1: Soft target construction. Firstly, the STF vector is employed to obtain the inter-class similarity of pollen via established template library (see Section 3.2 for detail introduction). We match pollen images with five categories of template images via Euclidean distance. For type i templates, the distances based on shape and texture are respectively obtained. Subsequently, these two types of distances are normalized and converted to similarity Sim_i^s and Sim_i^t by conversion functions Fun_s and Fun_t (as shown in Figs. 6 and 7, see Section 3.2 for details), respectively. Finally, we fuse Sim_i^s and Sim_i^t using Eq. (17) to get the total similarity (namely the value of the soft target on pollen category i).

$$p_i = w_1 * Sim_i^s + w_2 * Sim_i^t \tag{17}$$

where w_1 and w_2 are weights associated with each pollen feature and $w_1 + w_2 = 1$. The weights are set according to the experiment, $w_1 = 0.344$, $w_2 = 0.656$ in this paper.

Step 2: Shape-texture knowledge distillation. To guide the classification network to learn comprehensive pollen key features, we use soft targets to supervise the CNN. In detail, based on the input pollen image, we first obtain the fully connected outputs (denoted as z_i) of the backbone. Then the Softmax classifier converts z_i into a probability vector

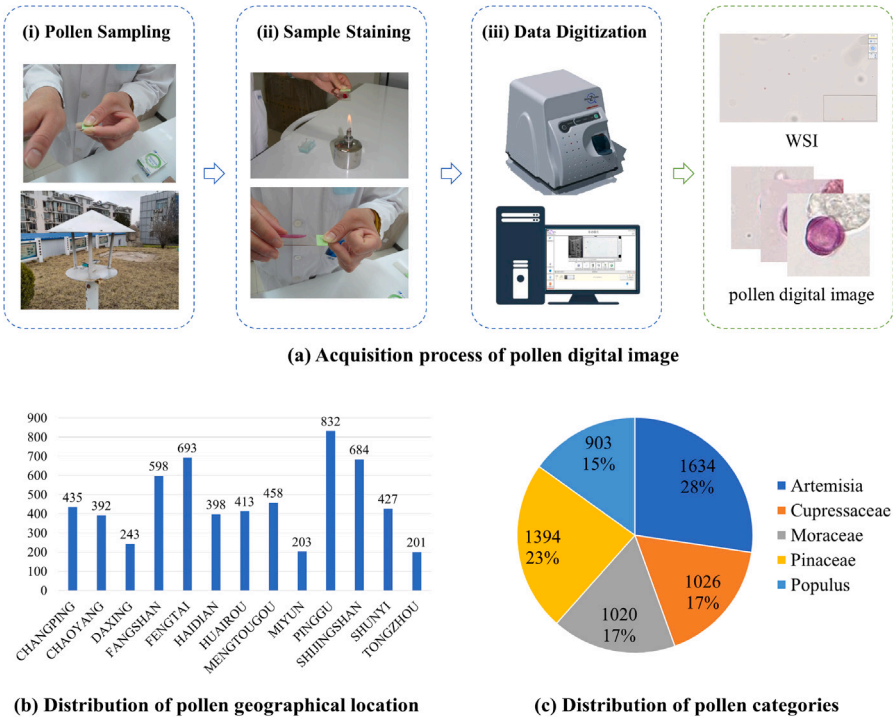


Fig. 8. Pollen digital image acquisition process and the data distribution of our dataset.

q_i^T indicating the predicted probability over each class as follows:

$$q_i^T = \frac{exp\left(z_i/T\right)}{\sum_j exp\left(z_i/T\right)} \tag{18}$$

where T is a temperature parameter that is normally set to 1 for classification tasks. While in knowledge distilling, a optimal value for T is important as it can generate soft probability distribution over classes. In this paper, T is set to 0.5 according to experiments. The soft target cross-entropy loss L_{soft} is introduced as follows:

$$L_{soft} = -\sum_i^N p_i log\left(q_i^T\right) \tag{19}$$

where N is the number of pollen categories. Hinton, Vinyals, Dean, et al. (2015) show that it is beneficial to combine ground truth labels with soft labels to supervise the classification network. Hence, we also calculate the hard loss as follows:

$$L_{hard} = -\sum_i^N c_i log\left(q_i^1\right) \tag{20}$$

where c_i is the ground truth on class i. q_i^1 is the softmax output with $T = 1$. The overall loss function is calculated as :

$$L = \beta L_{soft} + (1-\beta) L_{hard} \tag{21}$$

where β is a trade-off parameter in order to control the relative importance of these two loss terms ($\beta = 0.2$ in this paper).

3. Experiments and results

In this section, we first introduce the datasets and experimental settings in this paper. Experiments and results are then presented to validate the performance of our method in pollen classification.

3.1. Dataset

The distribution of pollen species is specific to different geographical regions. Collecting pollen samples in a specific area is important for pollen identification in that area. To realize the automatic classification of pollen in Beijing, China, we collect pollen grains from this area. With the strong support of Beijing Meteorological Service Center, we collect the five most allergenic pollen species to construct our pollen digital image dataset. As shown in Fig. 8(a), pollen digital image acquisition includes three steps: pollen sampling, sample staining, and data digitization. Specifically, the staff first use microscope slides to collect airborne pollen grains by gravity sedimentation. Then, an appropriate amount of Fuchsin is used to stain the sample to highlight the pollen characteristics. Finally, these slides are scanned with a NanoZoomer-SQ Slide Scanner to generate whole slide images (WSIs) in ndpi format. We process these WSIs to obtain pollen images with a size of 100 × 100 pixels.

Through the above methods, we obtain 5977 pollen digital images from Beijing (the geographical distribution of pollen images is shown in Fig. 8(b)). Our dataset includes five types of pollen with the strongest allergenicity in Beijing, namely Artemisia, Cupressaceae, Moraceae, Pinaceae, and Populus (the distribution of pollen categories is shown in Fig. 8(c)). Examples of each type of pollen image are shown in Fig. 9. During the pollen sampling phase, pollen grains and impurities in the air (such as animal debris, plant debris, and dust) are collected onto slides. In addition, it is easy to produce bubbles when the sample is stained and the cover glass is covered. Therefore, the digital pollen image contains not only pollen grains but also irrelevant targets (i.e., impurities and bubbles). In our experiments, 20 images are randomly selected from each category as template images (used in the template matching of the pollen classification module). For the remaining data, the ten-fold cross-validation is performed for evaluation purposes.

3.2. Experimental setup

Pollen Location Module Settings. The spatial window radius and the color window radius are both set to 15 in the mean shift filter. The Gaussian kernel size of the Gaussian blur is set to $(3, 3)$. The constant value in the image arithmetic operation is 1.5. The color threshold of image binarization includes low ($[150, 20, 20]$) and high ($[179, 255, 255]$)

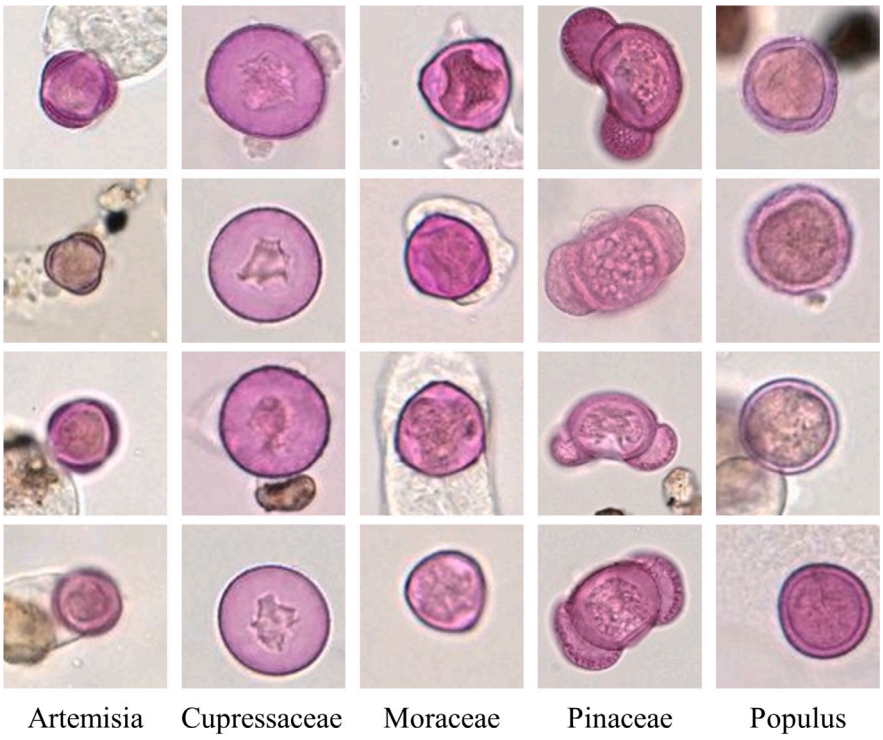


Fig. 9. Some examples of each allergic pollen class.

values. The pixel values between the low and high values are set to 255, while others are set to 0. The kernel sizes for closed and open operations are 30 × 30 and 20 × 20, respectively.

Shape-Texture Awareness Module Settings. In the *shape awareness*, the two thresholds of the Canny edge detection operator are set to 50 and 100, respectively. We set the hyperparameter n_t to 6 in the *texture awareness*.

Pollen Classification Module Settings. The ImageNet classification pre-trained model is used for weight initialization of the backbone network. For training the classification network, we optimize the network using SGD optimizer (Robbins & Monro, 1951) with batch of 16 images per step and an initial learning rate of $1e{-}4$. The weight decay factor is set to $1e{-}4$ and the momentum is 0.9. At each iteration, the loss of the model is recorded, and the loss of the neural network is assessed using an independent validation set. The maximum number of iterations is chosen to 100 epochs. Based on Early Stop, the training is stopped when the loss on the validation set no longer decreases. During training the CNN, we resize the pollen image and the STA map to 224 × 224 as input. The threshold α that controls the influence of the STA map in the *Attention Instruction* is set to 0.4. In the *Soft Target Supervision*, the temperature T is set to 0.5 and the β is set to 0.2.

Template Feature Vector Construction. Our template library contains five pollen types, each containing 20 template images. According to the shape-texture awareness module, each template image is represented by shape and texture feature vectors. We calculate the element-level mean of 20 shape/texture feature vectors to obtain the template STF vector representing the pollen class i. The shape and texture feature vectors of the class i template are respectively represented as: $Te^S = 1/20 \sum_{m=1}^{20} Te_m^S, Te^T = 1/20 \sum_{m=1}^{20} Te_m^T, i = 1, 2, \ldots, 5$, where m represents the mth image of the type i template.

Conversion Function Construction. The frequency distributions of Euclidean distances $d(Te_p^S, Te_j^S)$ and $d(Te_p^T, Te_j^T)$ are calculated, where p and j denote the template images of the same category. This process is performed on each type of pollen template. We collect the frequency distribution of all pollen categories to obtain a histogram $H(d)$. The conversion functions are three-segment functions on the distance interval $[0, +\infty)$. Segment 1: a constant function (the value is 1) in the interval $[0, \mathcal{U})$, where $\mathcal{U}$ is the mean of distances. Segment 2: a linear function that estimates the right-tail of the histogram on the interval $[\mathcal{U}, \mathcal{V})$, where $H(\mathcal{V}) = 0$. Segment 3: a constant function (the value is 0) on the interval $[\mathcal{V}, +\infty)$. Figs. 6 and 7 illustrate conversion functions (Fun_s and Fun_t) for the frequency distribution based on shape and texture distance, respectively.

Evaluation Metric. Following Battiato et al. (2021) and D'Orazio et al. (2021), we use accuracy (Acc), macro F1-score (M-F1), balanced accuracy (BAcc) and weighted F1-score (W-F1) to evaluate the performance of pollen classification models. The accuracy is the fraction of correctly classified samples. The macro F1-score computes the F1 for each label and returns their average. The balanced accuracy is defined as the average of recall obtained on each class. The weighted F1-score computes the F1 for each label and find their average weighted by the number of true instances for each label.

Implementation. The proposed model is implemented using the PyTorch framework with CUDA v11.0, and is coded in Python. All experiments are performed under an Ubuntu 18.04.5 LTS operating system on a machine with CPU Intel(R) Xeon(R) CPU E5-2609 v3 @ 1.90 GHz, GPU NVIDIA GeForce RTX 2080Ti and 62 GB RAM.

3.3. Evaluation on pollen classification

In this section, we compare the pollen image classification performance of our PK-DFL with other existing methods. Note that the models of other methods are retrained over the whole training set of our real-world dataset for a fair comparison. Table 1 shows the pollen identification performance of different methods. As we can see from the results, our proposed method is considerably effective, achieving 88.96%, 77.86%, 79.40% and 88.20% in terms of imbalanced accuracy, balanced accuracy, macro F1-score and weighted F1-score, respectively. Compared with other methods, our method achieves the best results on these metrics. Compared with using only VGG19 (our baseline), the imbalanced accuracy, balanced accuracy, macro F1-score and weighted F1-score are improved by 3.33%, 5.25%, 5.49% and 3.59% respectively. In addition, Table 2 shows the F1-score of every

Table 1
Comparison of classification performance of our PK-DFL with other pollen classification methods. The Training/Testing Time refers to the average training/testing time for each pollen image.

Method	Acc	BAcc	M-F1	W-F1	Train time (s)	Test time (s)
DenseNet121 (Huang, Liu, Van Der Maaten, & Weinberger, 2017)	84.10%	69.09%	72.06%	83.56%	**5.14e−1**	**1.37e−1**
VGG19 (Simonyan & Zisserman, 2014)	85.63%	72.61%	73.91%	84.61%	3.15	2.03e−1
ResNet18 (He, Zhang, Ren, & Sun, 2016)	81.10%	73.08%	72.94%	81.15%	1.98	1.51e−1
Gui et al. (2021)	80.34%	71.14%	70.17%	80.54%	3.69e+2	1.51
Trenta et al. (2021)	80.51%	66.97%	68.42%	79.45%	2.77e+2	8.31e−1
Ours	**88.96%**	**77.86%**	**79.40%**	**88.20%**	5.44	2.20e−1

Table 2
F1-score comparisons of each pollen category among our method and others. The category abbreviation consists of the first two letters of each type of pollen.

Method	Ar.	Cu.	Mo.	Pi.	Po.
DenseNet121 (Huang et al., 2017)	73.88%	**97.73%**	40.70%	**97.92%**	86.43%
VGG19 (Simonyan & Zisserman, 2014)	90.84%	90.84%	49.08%	91.50%	79.30%
ResNet18 (He et al., 2016)	86.62%	89.70%	48.73%	83.27%	77.78%
Gui et al. (2021)	80.43%	90.24%	44.55%	90.09%	77.11%
Trenta et al. (2021)	86.65%	87.27%	44.97%	86.13%	71.23%
Ours	**92.28%**	96.19%	**60.26%**	91.29%	**87.61%**

fine-grained pollen category. Our method achieves the highest F1-score in three pollen categories of Artemisia, Moraceae, and Populus, and obtains similar performance to the best F1-score in the other two pollen categories.

Our method has achieved remarkable performance in both overall and single category identification. We believe that the impressive results achieved by our model owe to the following factors. (1) We can accurately extract the shape and texture features of pollen grains and use them as prior knowledge to instruct the classification network. (2) The STA map is constructed to instruct the CNN to focus on critical areas of the pollen image. (3) The STF vector is calculated to supervise the CNN attending to comprehensive key features.

We quantitatively compare the computational time of the proposed model with other methods, as shown in Table 1. The average training and testing time for our PK-DFL is about 5.44 and 2.20e−1 s per pollen image, respectively. The time complexity of our model is lower than that of the pollen classification method proposed by Gui et al. (2021) and Trenta et al. (2021). We speculate that the key strategies of these studies (Cut Occlusion Augmentation and Test-Time Augmentation) multiply the number of images fed into the network and therefore the training and testing time is longer. Although the time complexity of our model is higher than that of the classic classifier (Densenet121 Huang et al., 2017, Vgg19 Simonyan & Zisserman, 2014, ResNet18 He et al., 2016), the proposed PK-DFL achieves the best results in terms of imbalanced accuracy, balanced accuracy, macro F1-score and weighted F1-score. For the plain VGG19 network, the training and testing time is about 3.15 and 2.03e−1 s per image, respectively. This shows that the time complexity of PK-DFL is similar to that of VGG19, but the PK-DFL can achieve a 3.33% performance gain (in balanced accuracy) compared with the VGG19. Importantly, the test time of 2.20e−1 s per image can fully meet the requirements of pollen identification in real-world scenarios.

3.4. Results of ablation experiments

In this section, we analyze the contribution of each component to the model's overall performance. The results are summarized in Tables 3 and 4. Table 3 shows the overall classification performance for all pollen categories. Table 4 shows the identification F1-score for each single fine-grained category of pollen.

Effectiveness of *Pollen location*. We first investigate the impact of *pollen location* for shape and texture extraction. To this end, we simply remove this module to predict pollen categories. As shown in Tables 3 and 4, compared with the backbone and our PK-DFL, the pollen classification performance is severely reduced. This is because the pollen location module produces pure pollen images guiding the shape-texture awareness module to extract pure pollen features rather than features of unrelated targets. This indicates that the guidance of accurate key features is crucial for obtaining better classification performance.

Table 3
The overall classification performance of different ablation schemes. PLM: pollen location module; ATI: Attention instruction; STS: Soft target supervision.

Method			Acc	BAcc	M-F1	W-F1
PLM	ATI	STS				
✗	✗	✗	85.63%	72.61%	73.91%	84.61%
✗	✓	✓	68.69%	59.34%	59.00%	69.80%
✓	✗	✓	87.39%	76.37%	77.85%	86.78%
✓	✓	✗	87.21%	75.80%	77.09%	86.46%
✓	✓	✓	**88.96%**	**77.86%**	**79.40%**	**88.20%**

Table 4
The F1-score on each pollen category among different ablation schemes. PLM: pollen location module; ATI: Attention instruction; STS: Soft target supervision. The category abbreviation consists of the first two letters of each type of pollen.

Method			Ar.	Cu.	Mo.	Pi.	Po.
PLM	ATI	STS					
✗	✗	✗	90.84%	90.84%	49.08%	91.50%	79.30%
✗	✓	✓	66.01%	82.11%	32.61%	79.67%	68.60%
✓	✗	✓	91.83%	93.39%	58.48%	92.17%	81.82%
✓	✓	✗	90.61%	93.83%	54.55%	**92.82%**	82.61%
✓	✓	✓	**92.28%**	**96.19%**	**60.26%**	91.29%	**87.61%**

Effectiveness of *Attention instruction*. Next, we assess the impact of *Attention instruction* on pollen classification performance. We remove this manner of domain knowledge guidance from the model to classify pollen categories. The results can be seen from Tables 3 and 4. When using *Attention instruction*, our method achieves 87.21% imbalanced accuracy, 75.80% balanced accuracy, 77.09% macro F1-score, and 86.46% weighted F1-score. Compared with using only the backbone, these indicators are improved by 1.58%, 3.19%, 3.18%, and 1.85% respectively. The classification F1-score of Cupressaceae, Moraceae, Pinaceae and Populus has also been greatly improved. And the F1-score of Artemisia is similar to that of the backbone. This is attributed to the fact that the *Attention instruction* enables the deep network to focus on critical regions instead of irrelevant ones. This result indicates the necessity of the *Attention instruction* for pollen identification.

Effectiveness of *Soft target supervision*. Finally, we remove this guidance manner to evaluate the impact of the *Soft target supervision* on pollen identification. As shown in Tables 3 and 4, the introduction

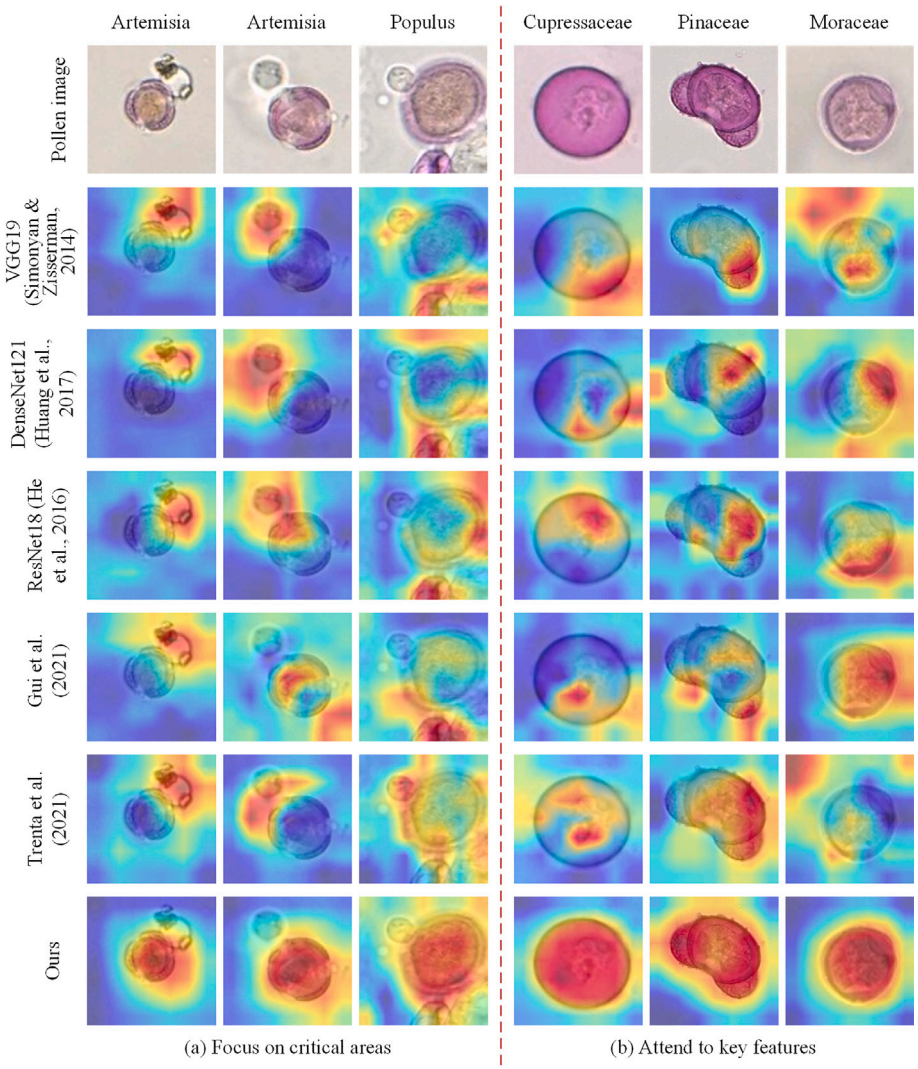


Fig. 10. The visualization results of different CNN models.

of this manner of guidance can improve the performance of pollen classification. For the identification performance of all categories, the imbalanced accuracy, balanced accuracy, macro F1-score and weighted F1-score are 87.39%, 76.37%, 77.85% and 86.78%, respectively. Compared with the backbone, the performance of the soft target supervised pollen classification model is improved in these indicators. For the classification performance of each single category, the identification F1-score increased 0.99%, 2.55%, 9.40%, 0.67%, and 2.52% respectively. This is because the *soft target supervision* guides the deep network to focus on comprehensive pollen key features. The above results show that *Soft target supervision* is effective for pollen classification.

4. Discussion

4.1. Model performance

Through extensive experiments, we see that our proposed model performs better than other classical pollen classification methods and each model component is highly effective. To intuitively demonstrate above quantitative results, we further use the Grad-CAM (Selvaraju et al., 2017) heat map approach to visually depict decisions made by the deep model. The heat map highlights important areas that the model emphasizes on the pollen image.

Effectiveness of the proposed model. Compared with other methods, the proposed model achieves satisfactory pollen classification results (see Tables 1 and 2). The visualization results shown in Fig. 10 further illustrate the effectiveness of the model. We observe that our method can focus on critical areas and attend to comprehensive key features. For example, in the first column, existing methods primarily focus on irrelevant information (i.e., impurities and bubbles), while our PK-DFL can accurately focus on the pollen grain. Moreover, in column 5, existing methods only focus on single or partial characteristics of Cupressaceae, while our model comprehensively attend to the shape and internal texture. Evidently, these visualization results further demonstrate the effectiveness of the proposed model in pollen identification.

Effectiveness of model components. As shown in Tables 3 and 4, *Attention instruction* and *Soft target supervision* are useful in that we combine the prior knowledge with CNNs to achieve better pollen classification results. In Fig. 11, we further illustrate the effectiveness of these two guidance manners through heat maps generated by Grad-CAM. As shown, the deep network with only *Attention instruction* can focus more on pollen areas, while features of pollen grains cannot be comprehensively concerned. Moreover, the deep network only guided by soft targets can attend to more comprehensive image features, but cannot accurately focus on pollen targets. When *Attention instruction* and *Soft target supervision* are incorporated, deep networks can not only focus on pollen regions, but also on comprehensive pollen features. In

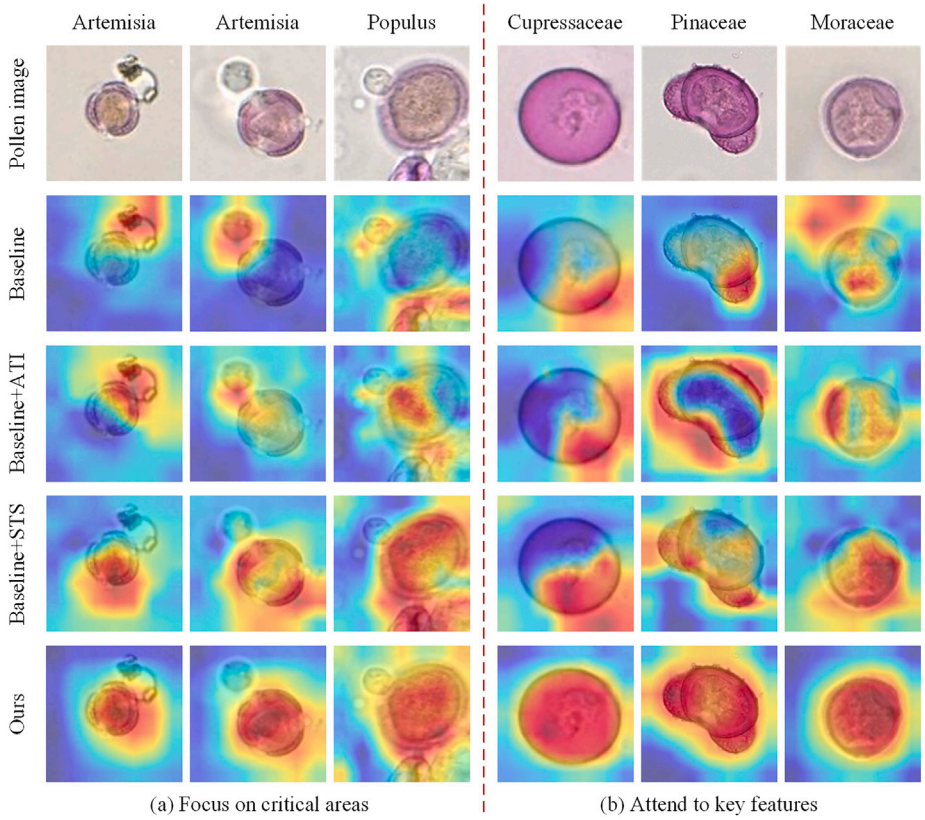


Fig. 11. The visualization results of different ablation schemes. ATI: Attention instruction; STS: Soft target supervision.

summary, our visualization results show that the *Attention instruction* and *Soft target supervision* are effective for pollen image classification.

4.2. Practical implication

We believe that our proposed method has great potential for practical applications. By imitating the expert's pollen identification manner, our model extracts the shape and texture features of pollen as prior knowledge to guide the deep feature learning of CNNs. Specifically, we construct STA maps to guide the deep network to focus on the key areas of the pollen image. We calculate STF vectors that are used to obtain category similarity by template matching. This information is transformed into soft targets to supervise the deep network focusing on more comprehensive pollen key features. Our model helps to improve performance of pollen image classification. The improved results can increase the accuracy of the pollen information. This helps allergic patients to assess disease risk and facilitates the prevention and treatment of pollinosis.

4.3. Limitation and future work

Although we have got good results in pollen classification, the following limitations must be acknowledged. (1) The shape and texture of pollen grains are used to guide the classification network in this paper. However, other prior features (e.g., apertures) are also crucial for pollen identification. More comprehensive prior knowledge helps to further improve the performance of pollen classification. (2) We only focus exclusively on five allergic pollen categories in Beijing, China. In fact, many other allergenic pollen species exist in Beijing and other areas. To improve model generalization ability, these categories could be considered in future studies. (3) We use two-dimensional images for the pollen classification task. However, pollen grains are three-dimensional objects, and their three-dimensional imaging contains more abundant pollen key information. Future studies should include three-dimensional imaging modalities.

Considering these limitations, the following improvements will be included in future work. (1) More prior knowledge will be considered to guide the deep network, thereby further improving pollen identification performance. For example, we will use rule-based methods to extract apertures of the pollen that will be used to assist the feature learning of the deep network. Moreover, we will consider not only visual cues but also additional information (e.g., the locations and dates for data collection). The potential of multimodal information will be fully explored. (2) We plan to expand the categories to include as many allergic pollen categories as possible for better allergic pollen monitoring. Besides, samples with non-allergenic pollen and samples with non-pollen (e.g., samples with sand of comparable sizes) will also be added to our dataset to enhance the generalization ability of the model. (3) We will use a NanoZoomer-SQ Slide Scanner to collect multilayer pollen data and build a corresponding classification model to capture the internal structure of pollen grains. A CNN will be employed to extract deep features of images in different layers. Considering the amount of information these layers carry and present is distinct, we will give different weights to these features for pollen classification.

5. Conclusion

We propose a new deep learning model PK-DFL, mimicking the basis and process of expert pollen identification, for OM pollen image classification in real-world scenarios. Our PK-DFL is composed of three modules: Pollen Location, Shape-texture Awareness and Pollen Classification. The pollen location module locates pollen grains based on color features to obtain pure pollen images that contribute to accurate prior feature extraction. Based on predefined feature extractors, the shape-texture awareness module extracts the shape and texture features

of pollen grains to construct the STA map and STF vector as prior knowledge to guide the next module. The pollen classification module uses a CNN to predict pollen categories under the guidance of prior knowledge. The STA map is weighted to the pollen image and the convolution feature map to instruct the deep network to focus on the critical areas of the pollen image (**Attention instruction**). We use STF vectors to obtain pollen intra-class similarity by template matching. This information is further converted into a soft target to supervise the CNN attending to comprehensive key features (**Soft target supervision**). Extensive experiments conducted on real-world datasets demonstrate that our PK-DFL significantly improves the performance of pollen identification. In future work, we will consider more prior knowledge to improve the pollen identification accuracy and further consider more allergenic pollen species to enhance the generalization ability of the model. Additionally, three-dimensional pollen imaging data will be collected and the internal structure of pollen grains will be captured.

CRediT authorship contribution statement

Jianqiang Li: Project administration, Supervision. **Wenxiu Cheng:** Conceptualization, Methodology, Writing – original draft. **Xi Xu:** Project administration, Supervision. **Linna Zhao:** Conceptualization, Methodology, Writing – original draft. **Suqin Liu:** Investigation, Data curation, validation. **Zhengkai Gao:** Investigation, Data curation, validation. **Caihua Ye:** Project administration, Supervision. **Huanling You:** Project administration, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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